

Muiz Jafri

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EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Engineering, Computer Engineering

2023–2028

- **Relevant Coursework:** Digital Systems, Advanced Algorithms, Object Oriented Programming, Operating Systems, Signals and Systems

EXPERIENCE

Software Engineer

Oct. 2025 – Present

Air Hawk Solutions

Oshawa, ON

- Built real-time Python pipelines on NVIDIA Jetson Nano by integrating LiDAR sensors and embedded communication protocols (I2C, UART, SPI), handling **1,000+ sensor data points per second** during autonomous system testing
- Worked on structuring and processing high-frequency sensor data for testing and debugging, helping reduce issue turnaround time by **30%**
- Built Python and Excel tools to track logs, sensor outputs, and system performance metrics, making it easier to spot issues during testing and simulation runs

Competitions Organizer

May 2025 – Present

MIST Toronto

Toronto, ON

- Built Excel and Google Sheets systems to manage scheduling, scoring, logistics, and participant tracking for **500+ participants** across multiple competitions
- Helped coordinate live events by updating schedules and resolving scoring or timing issues in real time under strict deadlines
- Worked with volunteers, judges, and organizers to keep competition workflows organized and running smoothly during events

Integration Camp Counsellor

May 2025 – Aug. 2025

City of Pickering

Pickering, ON

- Managed daily schedules, activity plans, and participant records for programs supporting **12+ participants**, ensuring consistent and structured delivery

TECHNICAL PROJECTS

Plant Classification Neural Network (In Progress) | *PyTorch, Computer Vision*

- Building a convolutional neural network (CNN) in PyTorch for plant image classification as part of learning deep learning fundamentals
- Working on dataset preprocessing and augmentation to improve model performance and generalization
- Learning and applying core concepts like backpropagation, gradient descent, and tensor operations through implementation

AI Typing Performance System | *Python, FastAPI, React, Hugging Face*

- Built an AI typing platform using Hugging Face models to generate dynamic typing prompts and improve user experience
- Developed FastAPI backend APIs to track user performance metrics like **WPM, accuracy, and error patterns** in real time
- Structured data flow between frontend, backend, and model outputs to keep performance tracking consistent across the system

Gold Price Predictor | *Python, Scikit-learn, Pandas, NumPy, Matplotlib*

- Built a machine learning model using Decision Tree Regression to predict gold prices using over **100 years of historical data**
- Created time-series features like lag values and rolling averages to better capture long-term trends and volatility
- Evaluated model performance using MAE, RMSE, and R^2 , and generated 12-month forecasts with visualizations

Real-Time Object Recognition System | *Python, OpenCV*

- Built a real-time computer vision system using OpenCV to process webcam video streams at up to **60 FPS**
- Used YOLO for object detection and BLIP for generating contextual labels from detected objects
- Structured outputs with timestamps and metadata for easier debugging and system analysis

TECHNICAL SKILLS

Programming: Python, C, C++, Java, JavaScript/TypeScript, SQL

Machine Learning / AI: PyTorch (learning), Scikit-learn, Pandas, NumPy, Hugging Face, YOLO, BLIP

Tools: Linux, Git, Jupyter Notebooks, FastAPI, Excel, Google Workspace, Microsoft Teams

Concepts: Data pipelines, real-time systems, computer vision, ML workflows, basic OS concepts, debugging, cross-team collaboration